TEAM QUANTA

INTERSTATE QUANTUM HACKATHON 2026

VARIATIONAL QUANTUM EIGENSOLVER FOR CHEMISTRY AND MATERIALS

Finding the ground state of stretched water

A variational quantum eigensolver study of circuit design, optimization, and hardware measurement

Asymmetric stretching of H_2O in a defined active space

PROJECT FOCUS

We study water at one asymmetric, stretched geometry. The study covers classical quantum chemistry references, variational circuits in noiseless simulation, a finite-shot Aer calculation, and separate measurements of the Hartree–Fock reference state on IBM hardware.

Group name	Team Quanta
Submission category	Variational Quantum Eigensolver for Chemistry and Materials
Main geometry	$R_1 = 0.96$ Å, $R_2 = 2.20$ Å, $\theta_{HOH} = 104.5°$
Electronic model	STO-3G basis, frozen-core CAS(8e,6o)
Qubit model	Parity mapping with two-qubit reduction, ten active qubits

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Project summary

Finding the ground state of stretched water • Team Quanta • Interstate Quantum Hackathon 2026

Abstract. We apply the variational quantum eigensolver (VQE) to water with one O–H bond stretched to 2.20 Å, the other held at 0.96 Å, and the H–O–H angle fixed at 104.5°. The model is STO-3G with a frozen-core CAS(8e,6o) active space mapped to ten qubits. Errors are measured against active-space CASCI, −74.860977098519 Ha, which an exact solve of the mapped Hamiltonian reproduces. In noiseless simulation the best paired-UCCSD parameter point lies 0.001520 mHa from CASCI, although all three trials stopped at their 1,500-evaluation budget. Adaptive fermionic VQE reached 3.473 mHa after eleven of sixteen requested cycles. A low-shot Aer run gave a standard error of 0.570414 Ha, and four IBM hardware jobs measured only the Hartree–Fock reference state. We report these kinds of evidence separately.

Keywords. variational quantum eigensolver, water, UCCSD, ADAPT, active space, IBM Quantum.

What we did

- Computed Hartree–Fock, CASCI, and full STO-3G FCI references with PySCF.
- Mapped the active space Hamiltonian to ten qubits with Qiskit Nature. An exact solve of the mapped problem agrees with CASCI.
- Optimized UCCSD, paired UCCSD (PUCCSD), and adaptive fermionic VQE in noiseless simulation, comparing energy error, parameter count, and CX gate count.
- Ran a three-point stretch scan and a CASCI against CASSCF check of the orbitals.
- Ran one low-shot Aer sampling calculation on a saved PUCCSD state.
- Measured the Hartree–Fock reference state on IBM hardware in four jobs.

Results at a glance

Part	Result	How to read it
Reference	CASCI −74.860977098519 Ha	The mapped exact solve gives the same energy. Full STO-3G FCI, −74.861014224915 Ha, is a different calculation and is not used to score VQE errors.
Best noiseless PUCCSD	−74.860975578262 Ha, error 0.001520 mHa	Three trials of 1,500 evaluations. All three hit the limit, so this is a good point reached on a fixed budget, not proof of convergence.
Resource pilot	PUCCSD, 3 repetitions: median 0.187 mHa, best 0.115 mHa, 90 parameters, 1,374 CX	Two restarts only. UCCSD with 1 repetition had best 0.064 mHa but median 7.193 mHa and 1,955 CX. Small samples, no ranking of ansätze.
Adaptive VQE	237.085 mHa down to 3.473 mHa after 11 cycles	Asked for 16 cycles, stopped as cycle 12 began. A separate 8-cycle run ended at 4.436 mHa. Neither reaches 1.6 mHa.
Finite-shot Aer	−74.071666227 Ha, standard error 0.570414 Ha	16 shots on each of 276 groups, 789.311 mHa from CASCI. Partial sums of −18 to −21 Ha in the log were unfinished sums, not errors.
IBM hardware	Four jobs, Hartree–Fock state only	Four X gates, no entangling gates. The optimized PUCCSD circuit was never run on hardware.

Interpretation rule. Noiseless VQE results are compared with active-space CASCI. The IBM jobs measure the Hartree–Fock reference, so they are compared with the ideal Hartree–Fock energy. Standard errors from the sampled and hardware runs are statistical only and leave out other systematic errors.

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1 Motivation

1.1 Why this track

We picked the chemistry and materials track because it asks a question we can actually test: does a quantum circuit get an electronic energy right, and what does it cost to get there? Chemistry gives us an exact classical answer to check against, so a result is either close to it or it is not. That kind of check suits a four-person student team with a few days and limited hardware time.

1.2 Why water, and why stretch it

The usual first VQE example is H_2, and it is too small to teach much. Water is only a little bigger, but it has a lone pair, two bonds, and enough orbitals that the choice of circuit starts to matter. We then stretch one O–H bond. Near the equilibrium geometry the Hartree–Fock picture is a decent description of water. As a bond stretches, that picture gets worse, and the circuit has to carry more of the correlation. That makes the problem a fair test of the ansätze we want to compare.

1.3 What we wanted to find out

We wanted answers to three practical questions:

- How close do UCCSD, paired UCCSD, and adaptive circuits get to the classical reference, and how many parameters and CX gates do they need?
- How much do optimizer choice and evaluation budget change the answer?
- What happens to an energy estimate when we move from an exact simulator to a few shots, and then to a real IBM device?

We were careful from the start to keep these three kinds of evidence separate. A noiseless simulation, a finite-shot simulation, and a hardware job answer different questions, and the report treats them that way.

2 The problem in everyday language

A molecule is a set of electrons that are attracted to the nuclei and repelled by each other. Its ground state is the lowest-energy arrangement the electronic Hamiltonian allows. Finding that state is one of the main jobs of computational chemistry.

Classical programs handle small molecules well. The trouble is that the number of possible

electron configurations grows very quickly as orbitals are added. A quantum computer stores the electronic state in qubits instead. VQE splits the work between the two machines: a circuit prepares a trial state, measurements estimate its energy, and a classical optimizer adjusts the circuit parameters to push that estimate down.

Water makes a good test case because it has more electronic structure than the simplest diatomic examples. We stretch one O–H bond and hold the other bond and the bond angle fixed. That gives a controlled change in geometry, and it shows how well each circuit copes with the extra correlation that stretching brings in.

2.1 Research question

The question we set out to answer is:

> For asymmetric stretching of H_2O in a fixed STO-3G active space, how accurately do selected VQE circuits reproduce a classical CASCI reference, and what parameter, gate, optimization, and measurement costs come with those results?

The question has a simulation half and a hardware half. The noiseless simulations test how accurate each ansatz is against an active-space reference. The hardware jobs test whether a chosen state and observable can be compiled, laid out, submitted, and measured on an IBM device. The two halves are linked engineering steps, but they are separate scientific results.

2.2 Why stretch one bond?

Hartree–Fock describes a molecule with one dominant electronic configuration. In a stretched geometry other configurations matter more, so a variational circuit has to build a more complicated superposition to get near the ground state.

The main geometry is

$$R_1 = 0.96\,\text{Å}, \qquad R_2 = 2.20\,\text{Å}, \qquad \theta_{\text{HOH}} = 104.5^\circ. \tag{1}$$

The oxygen atom sits at the origin. The first hydrogen is on the positive z axis and the second lies in the xz plane:

$$\mathbf{R}_{\text{O}} = (0, 0, 0), \tag{2}$$
$$\mathbf{R}_{\text{H}_1} = (0, 0, R_1), \tag{3}$$
$$\mathbf{R}_{\text{H}_2} = (R_2 \sin\theta,\ 0,\ R_2 \cos\theta)\,. \tag{4}$$

At $R_2 = 2.20$ Å, the second hydrogen is at about $(2.12992480883, 0, -0.55083600892)$ Å.

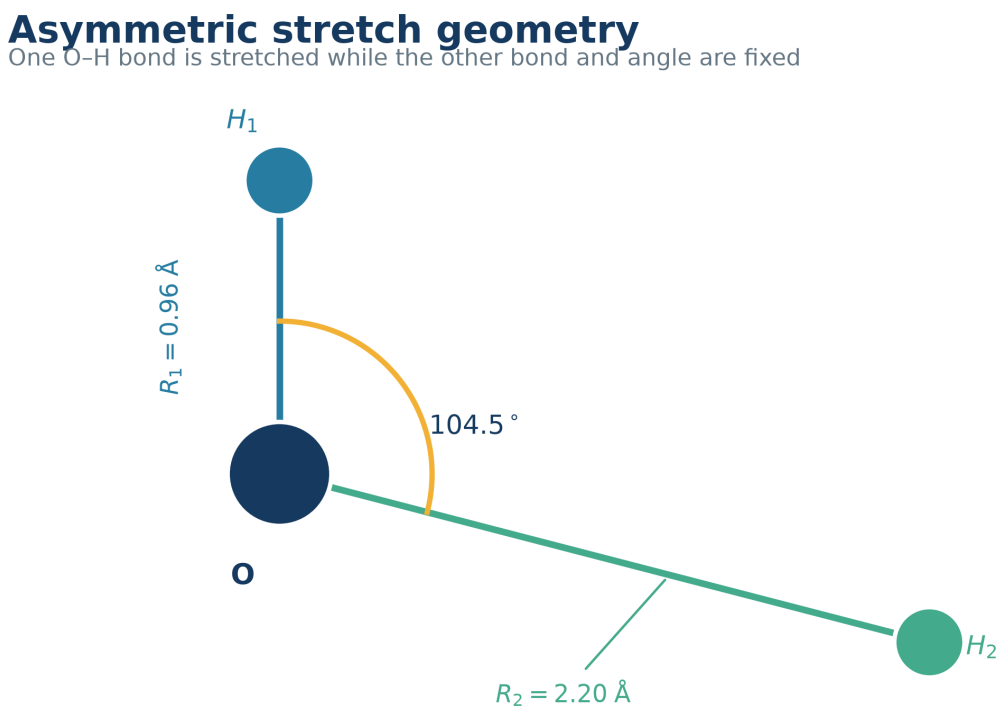


Figure 1: *Water geometry used for the asymmetric stretch.*

3 From a molecule to a qubit Hamiltonian

3.1 Basis set and active space

STO-3G is a minimal basis set. It keeps the electronic structure calculation small enough for a controlled VQE study. It is far from a converged description of water, so we do not use it to make claims about experimental accuracy.

The full water calculation has ten electrons. In the quantum calculation we freeze the oxygen 1*s* core orbital, which leaves eight active electrons in six spatial orbitals. This is written CAS(8e,6o). Every spatial orbital has two spin orbitals, so the active problem starts with twelve spin orbitals.

Freezing the core cuts down the number of active degrees of freedom. The core still contributes through the effective Hamiltonian and the constant energy terms. What we solve is a defined, reduced electronic problem and not the full all-electron one.

3.2 Electronic Hamiltonian

Before any mapping to qubits, the electronic Hamiltonian is written with fermionic creation and annihilation operators:

$$\hat{H}_{\text{elec}} = \sum_{pq} h_{pq} \hat{a}_p^\dagger \hat{a}_q + \frac{1}{2} \sum_{pqrs} g_{pqrs} \hat{a}_p^\dagger \hat{a}_q^\dagger \hat{a}_s \hat{a}_r + E_{\text{nuc}}. \tag{5}$$

Here h_{pq} are the one-electron integrals, g_{pqrs} are the two-electron integrals in the convention of the chemistry software, and E_{nuc} is the nuclear repulsion energy. The operators $\hat{a}_p^\dagger$ and $\hat{a}_p$ create and remove electrons in spin orbitals.

Once the inactive core is folded in, the active Hamiltonian takes the form

$$\hat{H}_{\text{active}} = E_{\text{inactive}} + \sum_{pq \in \mathcal{A}} h_{pq}^{\text{eff}} \hat{a}_p^\dagger \hat{a}_q + \frac{1}{2} \sum_{pqrs \in \mathcal{A}} g_{pqrs} \hat{a}_p^\dagger \hat{a}_q^\dagger \hat{a}_s \hat{a}_r, \tag{6}$$

where $\mathcal{A}$ is the set of active spin orbitals and E_{inactive} holds the frozen-core and nuclear contributions.

3.3 Mapping fermionic occupations to qubits

The parity mapping stores fermionic occupation information in qubits. We then apply a two-qubit reduction that uses particle-number and spin-parity symmetry. The active space Hamiltonian ends up acting on ten qubits.

The mapped Hamiltonian is a weighted sum of Pauli strings:

$$\hat{H}_{\text{q}} = E_{\text{offset}} \hat{I} + \sum_{j=1}^{M} c_j \hat{P}_j, \qquad \hat{P}_j \in \{I, X, Y, Z\}^{\otimes 10}. \tag{7}$$

Each c_j is a real coefficient in Hartree, and each $\hat{P}_j$ is a tensor product of single-qubit Pauli operators. The constant offset is added back when a mapped expectation value is converted into a total molecular energy.

3.4 Reference energies

A VQE energy only means something next to a reference for the same Hamiltonian and electronic model. Table 1 lists the references at the main geometry.

The mapped exact solve matches CASCI at the precision we report. That tells us the active space Hamiltonian, the mapping, the reduction, and the energy offset all describe the CASCI problem we meant to solve.

The Hartree–Fock to CASCI gap is

Table 1: *Reference energies at $R_2 = 2.20$ Å. The CASCI and full FCI values refer to different electronic problems.*

Calculation	Energy (Ha)	What it represents
Hartree–Fock	−74.623892380736	Mean-field reference determinant
Frozen-core CASCI(8e,6o)	−74.860977098519	Main active-space VQE reference
Mapped active-space exact solve	−74.860977098519	Independent check of the mapped active-space problem
Full STO-3G FCI	−74.861014224915	Full configuration interaction in the full STO-3G orbital basis

$$E_{\mathrm{HF}} - E_{\mathrm{CASCI}} = (-74.623892380736) - (-74.860977098519)\,\mathrm{Ha} \tag{8}$$
$$= 0.237084717783\,\mathrm{Ha} \tag{9}$$
$$= 237.084718\,\mathrm{mHa}. \tag{10}$$

This is the correlation energy that Hartree–Fock leaves out in this active-space comparison. It is a property of the reference calculation and not an energy gain measured on hardware.

The full STO-3G FCI value sits slightly below the frozen-core CASCI value because it covers a larger configuration space. We list it for context only. It plays no part in the active-space VQE errors.

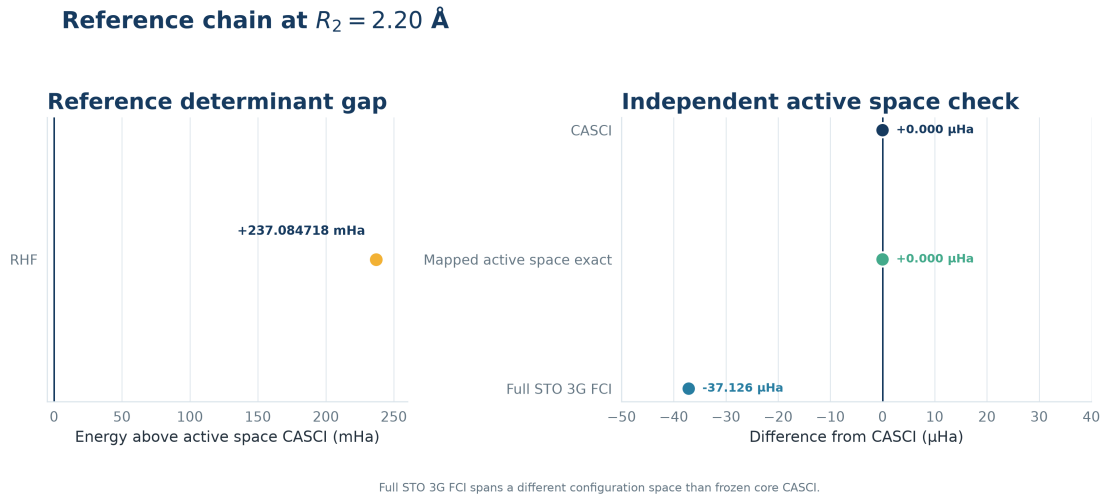


Figure 2: *Classical reference energies at the main geometry, shown relative to CASCI.*

4 How VQE works

4.1 The central idea

VQE starts from a prepared reference state, which here is the Hartree–Fock determinant. A parameterized circuit transforms it. A quantum processor or simulator estimates the energy of the result, and a classical optimizer changes the parameters and asks for another estimate.

The loop ends for one of several reasons: numerical convergence, an iteration limit, an evaluation budget, or a cap on how many times an adaptive circuit may grow. These are different outcomes. A run that stops at its budget has not converged unless its convergence criteria were also met.

4.2 Variational principle

For a normalized state $|\psi(\boldsymbol{\theta})\rangle$, the variational energy is

$$E(\boldsymbol{\theta}) = \langle\psi(\boldsymbol{\theta})|\hat{H}|\psi(\boldsymbol{\theta})\rangle. \tag{11}$$

For the ground-state energy E_0 of the same Hamiltonian,

$$E(\boldsymbol{\theta}) \geq E_0. \tag{12}$$

The trial state comes from a unitary circuit acting on the reference:

$$|\psi(\boldsymbol{\theta})\rangle = U(\boldsymbol{\theta})|\Phi_{\mathrm{HF}}\rangle. \tag{13}$$

The ansatz decides which states the circuit can reach. The optimizer decides which of those states we actually visit, given a starting point and a budget.

4.3 Energy estimation from Pauli measurements

Using the mapped Hamiltonian in Equation 7,

$$E(\boldsymbol{\theta}) = E_{\mathrm{offset}} + \sum_{j=1}^{M} c_j \langle\hat{P}_j\rangle_{\boldsymbol{\theta}}. \tag{14}$$

To estimate one Pauli expectation value we prepare the state and measure the right qubits. Terms that share a compatible measurement basis can be grouped. Grouping cuts the number of distinct circuit settings, but every group still needs repeated shots.

The number of groups is not the number of shots. A Hamiltonian with hundreds of groups needs a lot of state preparations even if each group only gets a few shots.

When the energy is built from many group estimates, its statistical uncertainty depends on the variance of each group and on the coefficients. A small shot count can therefore give a large uncertainty, especially when many groups contribute.

5 Circuit ansätze

5.1 UCCSD

Unitary coupled cluster with singles and doubles uses single and double electron excitation operators. In the usual notation,

$$U_{\mathrm{UCCSD}}(\boldsymbol{\theta}) = \exp\left[\hat{T}_1(\boldsymbol{\theta}) + \hat{T}_2(\boldsymbol{\theta}) - \hat{T}_1^\dagger(\boldsymbol{\theta}) - \hat{T}_2^\dagger(\boldsymbol{\theta})\right]. \tag{15}$$

A single excitation moves one electron between orbitals and a double excitation moves two. Applying the exponential to the Hartree–Fock determinant gives a correlated trial state.

The mapped excitation operators then have to be compiled into native gates. The final gate count depends on the mapping, the operator ordering, how the circuit is synthesized, the transpiler optimization level, and the hardware basis. The CX counts in this report are the optimization-level-3 transpilation counts from the noiseless resource pilot.

5.2 Paired UCCSD

Paired UCCSD limits the excitations so that electron pairs mostly move together. That can cut the number of independent parameters and gates. How expressive the circuit is depends on the number of repetitions, or layers.

A shallow paired circuit is cheaper, but it may not reach the energy of a more expressive one. Each extra repetition adds parameters and gates. Our results give one numerical example of that tradeoff, for this geometry and this active space.

5.3 Adaptive fermionic VQE

ADAPT builds its circuit one operator at a time. In each macrocycle it computes gradients for a pool of candidate fermionic excitations, picks the one with the largest gradient magnitude, appends it to the circuit, and then reoptimizes every parameter already in the circuit.

In schematic form, for an operator A_k in the pool, the energy derivative at the current state is

$$\left.\frac{\partial E}{\partial \theta_k}\right|_{\theta_k=0} = \langle \psi | [\hat{H}, A_k] | \psi \rangle, \tag{16}$$

where the exact convention depends on the anti-Hermitian generator the implementation uses. The operator that gets selected is the one with the largest absolute gradient under that implementation.

ADAPT can build a circuit tailored to the current state, but it has costs of its own. The pool gradients have to be evaluated, and all active parameters are reoptimized after each growth step. The number of macrocycles alone therefore says little about the total work done.

6 Calculation workflow and code

6.1 Workflow

We kept the chemistry references, the noiseless variational calculations, the finite-shot simulation, and the hardware measurements apart from each other. The steps were:

1. Define the molecular coordinates, charge, spin, and STO-3G basis.
2. Run PySCF to get molecular orbitals and the Hartree–Fock energy.
3. Freeze the oxygen 1*s* core and build the CAS(8e,6o) active space.
4. Compute CASCI as the active-space reference and full STO-3G FCI as a separate all-electron reference.
5. Map the active space Hamiltonian with the parity mapping and two-qubit reduction.
6. Solve the mapped problem exactly and compare its energy with CASCI.
7. Build the Hartree–Fock initial state and the chosen VQE ansatz.
8. Optimize the parameters in noiseless simulation and record energies, restart information, and circuit resources.
9. Run a separate finite-shot Aer calculation on a saved ansatz state.
10. Submit separate IBM jobs that prepare the Hartree–Fock reference state.
11. Compare each result with the right reference and report its uncertainty and how the run stopped.

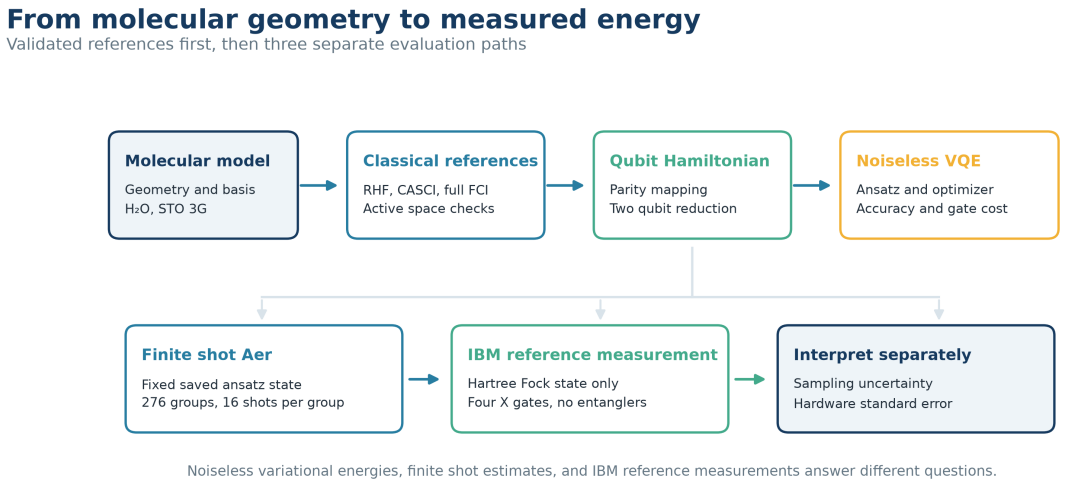


Figure 3: *Workflow from molecular geometry to reference checks, simulation, and hardware measurement.*

6.2 Code layout

All code sits in a public GitHub repository. The runnable notebook is `notebooks/h2o_asymmetric_stretch.ipynb`, and the maintained VQE implementation is `src/h2o_vqe/engine.py`. The first cell of the notebook installs the packages it needs, in the form `%pip install numpy scipy matplotlib ipywidgets pylatexenc qiskit pyscf qiskit_nature`.

The notebook walks through the steps above. API keys and tokens are never stored in the repository or in this report.

7 Noiseless simulation results

7.1 Ansatz resource pilot

The resource pilot used two restarts for each configuration at $R_2 = 2.20$ Å. Errors are absolute differences from the active-space CASCI energy, and CX counts are the optimization-level-3 transpiled values.

The median error fell as we added repetitions to the paired ansatz. One repetition gave a median of 8.524 mHa, two gave 2.876 mHa, and three gave 0.187 mHa.

The gain was paid for in parameters and CX gates, which grew in proportion. One repetition used 30 parameters and 458 CX gates, and three repetitions used 90 parameters and 1,374 CX gates. UCCSD with one repetition reached a best error of 0.064 mHa, but its median was higher because the two trials landed far apart.

We read this narrowly. With this implementation and this pilot setup, the deeper paired circuit

Table 2: *Noiseless ansatz resource pilot. These are small pilot samples, not statistical guarantees.*

Ansatz	Parameters	CX gates	Median error (mHa)	Best error (mHa)	Trials within 1.6 mHa
UCCSD, one repetition	92	1,955	7.193	0.064	1 of 2
PUCCSD, one repetition	30	458	8.524	8.518	0 of 2
PUCCSD, two repetitions	60	916	2.876	2.517	0 of 2
PUCCSD, three repetitions	90	1,374	0.187	0.115	2 of 2

reached a lower observed error at a higher circuit cost. Two restarts cannot show that the paired ansatz beats UCCSD or any other family in general.

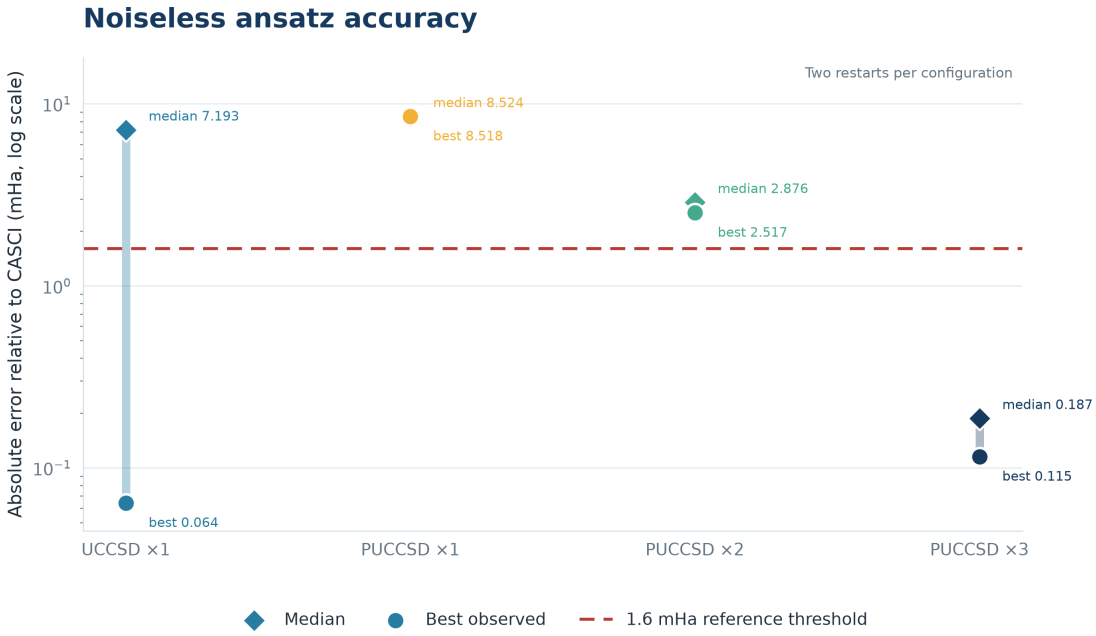


Figure 4: *Noiseless ansatz accuracy in the resource pilot. Median and best of two restarts, logarithmic error axis.*

7.2 Accuracy and circuit resource tradeoff

Parameter count and CX count measure different things. Parameters set the size of the classical search space. CX gates drive circuit depth and with it the exposure to hardware error. Neither number alone tells us how a noisy hardware run would turn out.

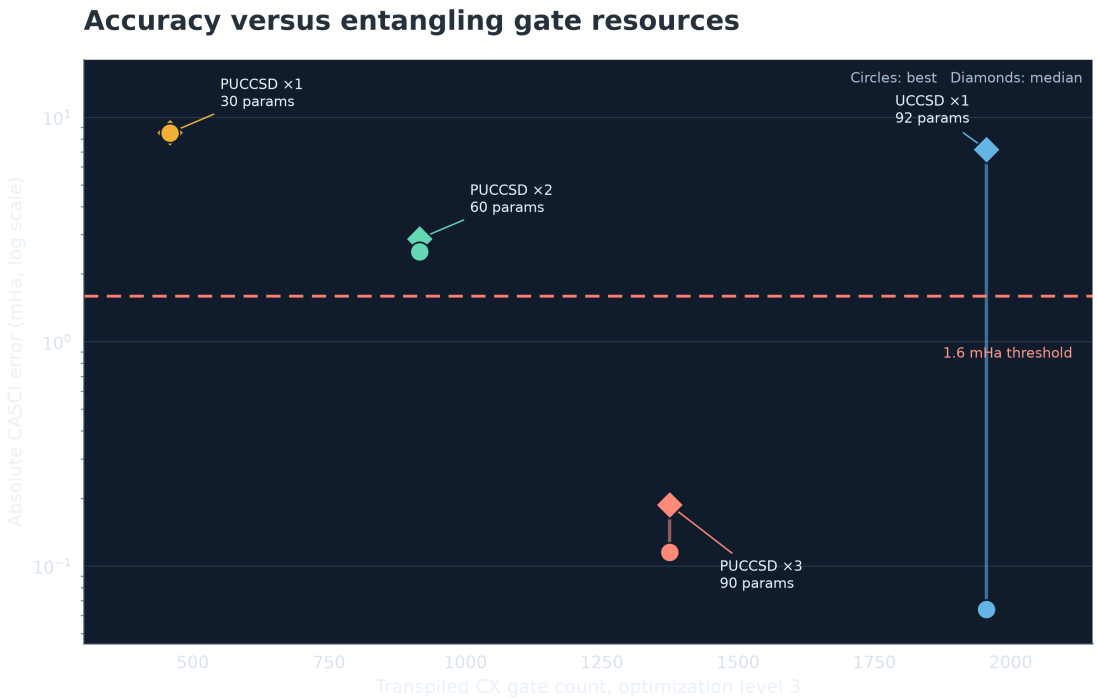


Figure 5: *Observed noiseless accuracy against transpiled entangling-gate count.*

7.3 Separate three-repetition optimization trials

A separate three-trial L-BFGS-B run gave each trial a budget of 1,500 objective evaluations. The terminal output reported these energies:

Table 3: *Three high-accuracy PUCCSD three-repetition trials. All reached the stated evaluation budget.*

Trial	Energy (Ha)	Absolute error (mHa)	Evaluations
0	−74.860975578262	0.001520	1,500
1	−74.860975150111	0.001948	1,500
2	−74.860961249707	0.015849	1,500

The best trial sits 0.001520 mHa from CASCI, far under the 1.6 mHa chemical-accuracy threshold for this active-space reference. Every trial used up its evaluation budget, though. These are good parameter points from budget-limited runs. The output does not show that the optimizer converged or that it found the global minimum of the ansatz.

A separate, completed SLSQP run reached −74.860885521062 Ha, about 0.0916 mHa above CASCI. It is also a noiseless result, and it is distinct from the three-trial L-BFGS-B run.



Figure 6: *Three reported PUCCSD three-repetition L-BFGS-B trials, each stopped at 1,500 evaluations.*

7.4 Asymmetric stretch scan

A warm-start scan used PUCCSD with three repetitions at three values of the stretched bond length. The other O–H distance stayed at 0.96 Å and the angle stayed at 104.5°.

Table 4: *Three-point asymmetric stretch scan with PUCCSD and three repetitions.*

R_2 (Å)	RHF (Ha)	CASCI (Ha)	PUCCSD energy (Ha)	Error (mHa)
1.80	−74.725988364448	−74.887678538893	−74.887280139717	0.398
2.20	−74.623892380736	−74.860977098519	−74.860687456423	0.290
2.60	−74.559694580830	−74.853880734062	−74.853326368035	0.554

All three VQE energies lie within 0.554 mHa of their CASCI values. This shows the workflow running at three geometries. It does not give a smooth, resolved potential energy curve, and the figure draws only the three computed points.

7.5 Orbital reference audit

The orbital audit compared CASCI with CASSCF at five values of R_2. The restricted Hartree–Fock solution was reported as externally unstable at every point, and CASSCF converged at all five.

CASSCF is lower than CASCI by roughly 0.022 to 0.026 mHa, so relaxing the orbitals changed the energy only a little here. We kept the RHF stability label as the audit reported it. How to

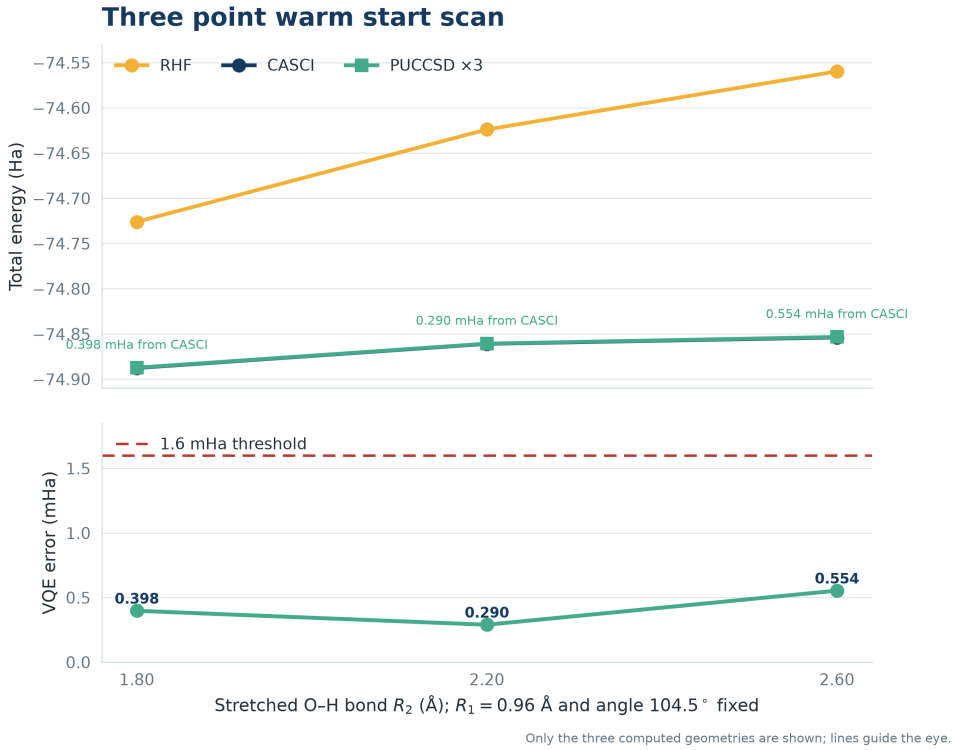


Figure 7: *Three calculated points along the asymmetric stretch. Panel (a) shows energies and panel (b) the PUCCSD error.*

Table 5: *CASCI and CASSCF reference audit across the asymmetric stretch.*

R_2 (Å)	CASCI (Ha)	CASSCF (Ha)	Difference (mHa)	RHF stability
1.80	−74.8876785389	−74.8877042972	−0.026	Externally unstable
2.00	−74.8704028691	−74.8704270924	−0.024	Externally unstable
2.20	−74.8609770985	−74.8610001275	−0.023	Externally unstable
2.40	−74.8562001454	−74.8562223635	−0.022	Externally unstable
2.60	−74.8538807341	−74.8539024560	−0.022	Externally unstable

read it depends on the stability procedure the calculation used.

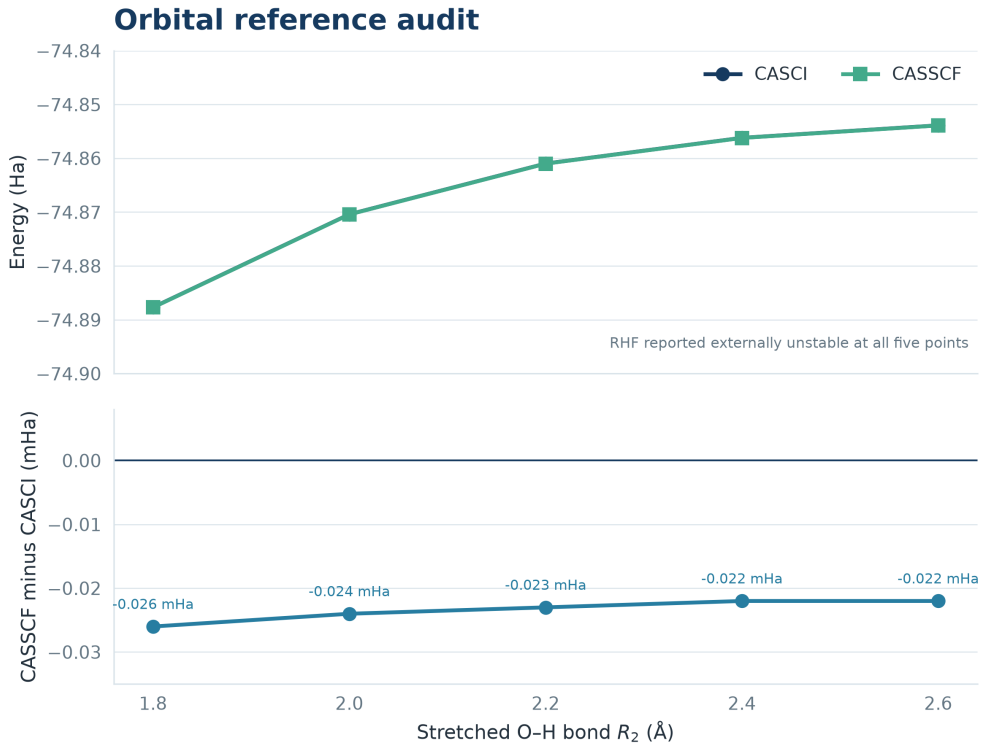


Figure 8: *Orbital reference audit at five asymmetric stretch geometries.*

8 Adaptive fermionic VQE

We have two ADAPT records. One run completed eight macrocycles. A later run asked for sixteen, completed eleven, and was stopped at the start of cycle twelve. The two should not be merged, and neither should be called a completed sixteen-cycle result.

8.1 Completed eight-cycle run

The eight-cycle run finished at −74.856541590 Ha, 4.436 mHa from CASCI. That is still above the 1.6 mHa threshold.

8.2 Later run requested for sixteen cycles

The later run was set up for sixteen macrocycles. The pasted log has completed results through cycle eleven, and then shows the program being stopped as cycle twelve began.

Table 6: *Final recorded energy after each completed macrocycle in the later ADAPT run.*

Completed cycle	Energy (Ha)	Absolute error (mHa)
1	−74.833020889	27.956
2	−74.847177291	13.800
3	−74.849676468	11.301
4	−74.852336686	8.640
5	−74.854687517	6.290
6	−74.855144209	5.833
7	−74.855814459	5.163
8	−74.856541590	4.436
9	−74.856845948	4.131
10	−74.857182481	3.795
11	−74.857504538	3.473

The recorded error fell from 237.085 mHa at the starting Hartree–Fock state to 3.473 mHa after eleven cycles. Between cycles eight and eleven it improved by 0.963 mHa. At the last recorded cycle it was still above 1.6 mHa.

The log contains no completed energy for cycle twelve, so this run is incomplete. The sixteen cycles we asked for must not be reported as the number completed.

8.3 How to read ADAPT cost

Every macrocycle needs gradients for all operators in the pool. After an operator is chosen, the circuit grows and all active parameters are optimized again. The count of VQE objective evaluations leaves out the cost of those pool gradients. So the ADAPT curve shows the energy reached at each recorded cycle. It does not give a full wall-clock or measurement-cost comparison with the fixed ansätze.

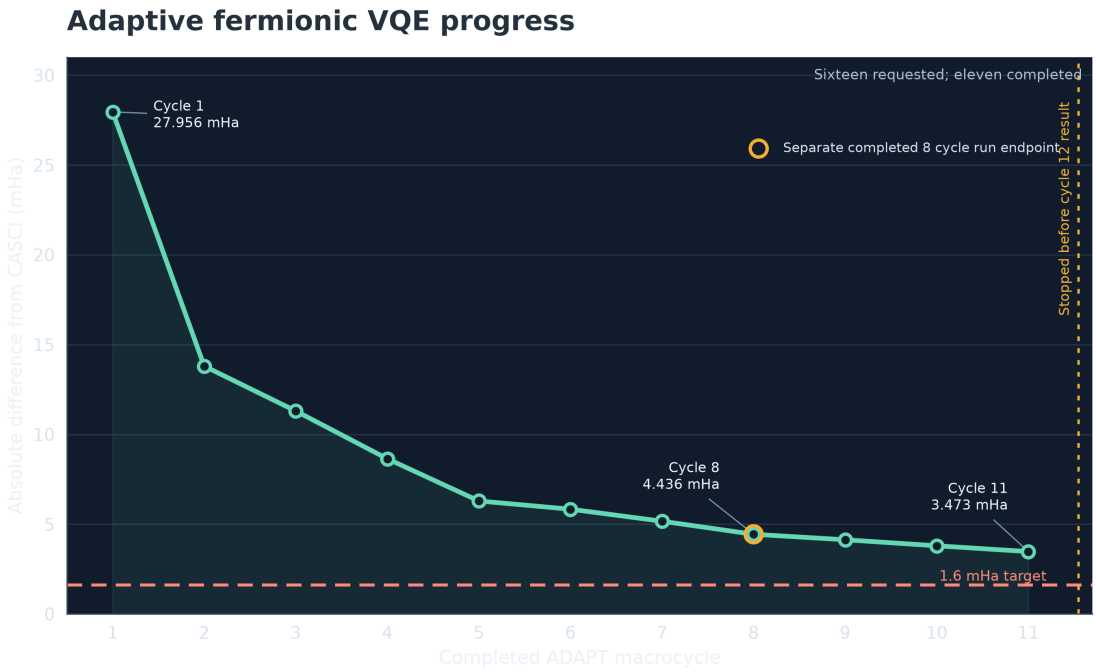


Figure 9: *Recorded ADAPT energy error over the eleven completed macrocycles.*

9 Optimizer experiments

We tried SLSQP, L-BFGS-B, COBYLA, Powell, and SPSA in several configurations. What an optimizer does depends on the ansatz, the starting parameters, the step-size settings, the stopping rule, and the evaluation budget. A short run that ends at a poor point tells us about that setting and that budget. It does not show that the whole optimizer family is ineffective.

The systematic held-out comparison used the one-repetition PUCCSD ansatz. Its median absolute errors were:

These medians do not give a universal ranking. They also say little about PUCCSD with three repetitions, which has three times as many parameters. A fuller comparison would list the per-run values, evaluation counts, and stopping statuses, if those records are available.

Some early pilot runs of COBYLA and SPSA ended above the Hartree–Fock starting energy. Those runs had short budgets or settings we had not tuned. They are unsuccessful runs under those conditions and say nothing against gradient-free optimization as such.

Table 7: *Held-out optimizer comparison for PUCCSD with one repetition. The ranking applies only to this ansatz and these study settings.*

Optimizer	Median error (mHa)	Scope
L-BFGS-B	8.518	One-repetition PUCCSD held-out runs
SLSQP	8.534	One-repetition PUCCSD held-out runs
COBYLA	9.574	One-repetition PUCCSD held-out runs
SPSA	106.586	One-repetition PUCCSD held-out runs
Powell	273.682	One-repetition PUCCSD held-out runs

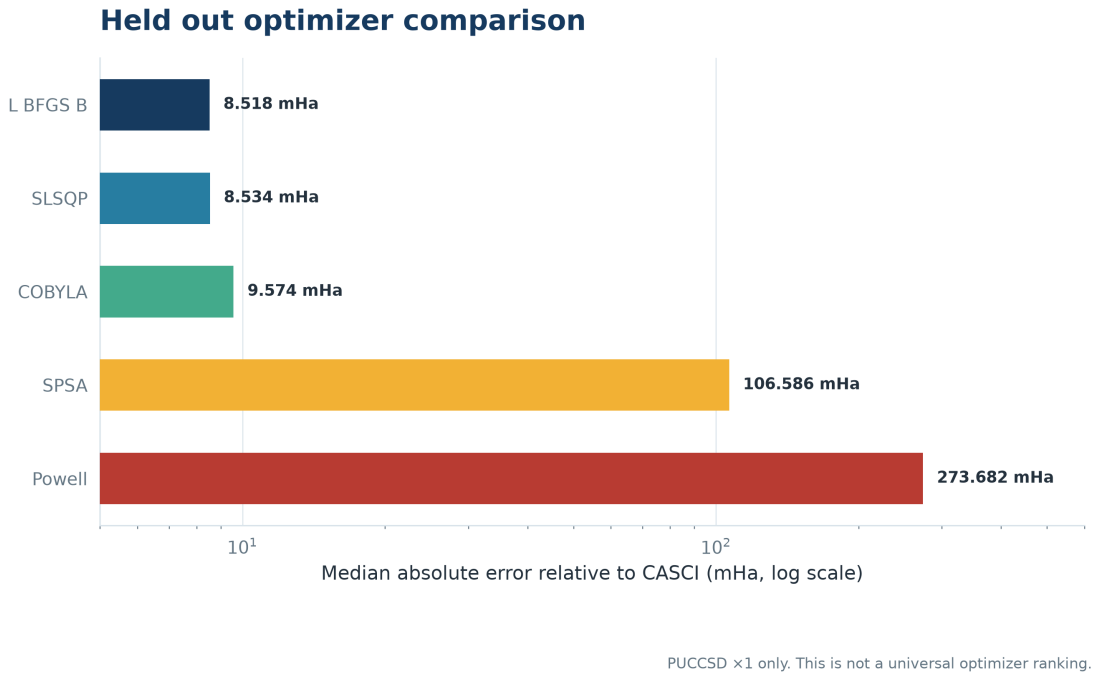


Figure 10: *Median error by optimizer for the one-repetition PUCCSD study.*

10 Finite-shot Aer simulation

10.1 Purpose and setup

A separate Aer calculation evaluated a saved one-repetition PUCCSD state with finite-shot sampling. The log reports 276 qubit-wise commuting measurement groups, sixteen shots per group, one seed, and 4,416 nominal shots in total.

An exact statevector check of the same prepared state gave −74.852459541 Ha, 8.518 mHa from CASCI. The sampled estimate came out very different from that.

10.2 Completed sampled estimate

Table 8: *Completed low-shot Aer estimate for a fixed PUCCSD one-repetition state.*

Quantity	Reported value	Meaning
Ansatz	PUCCSD, one repetition	Fixed saved noiseless parameter vector
Measurement groups	276	Qubit-wise commuting groups
Shots per group	16	4,416 nominal shots total
Seed	11	One random seed recorded
Exact statevector energy	−74.852459541 Ha	Ideal energy of the prepared ansatz state
Sampled total energy	−74.071666227 Ha	Completed finite-shot energy estimate
Sampling standard error	0.570414 Ha	570.414 mHa
Absolute difference from CASCI	789.311 mHa	Much larger than 1.6 mHa
Elapsed time	1,079.6 seconds	Recorded run time

The uncertainty is far larger than the chemical-accuracy threshold. The calculation finished, but it does not give an accurate ground-state energy. It shows what a very small shot budget does when the Hamiltonian has many measurement groups.

While it processes groups, the log prints partial Hamiltonian sums. Values near −18 and −21 are in Hartree and are labelled as incomplete. They are not millihartree errors. The full energy exists only once all 276 group contributions and the constant offset are added together.

The pasted log shows a noisy Aer sampling procedure but does not give every detail of the simulator noise channel. We therefore call this the recorded finite-shot Aer result, and we do not name a physical-device noise model unless the run settings confirm one.

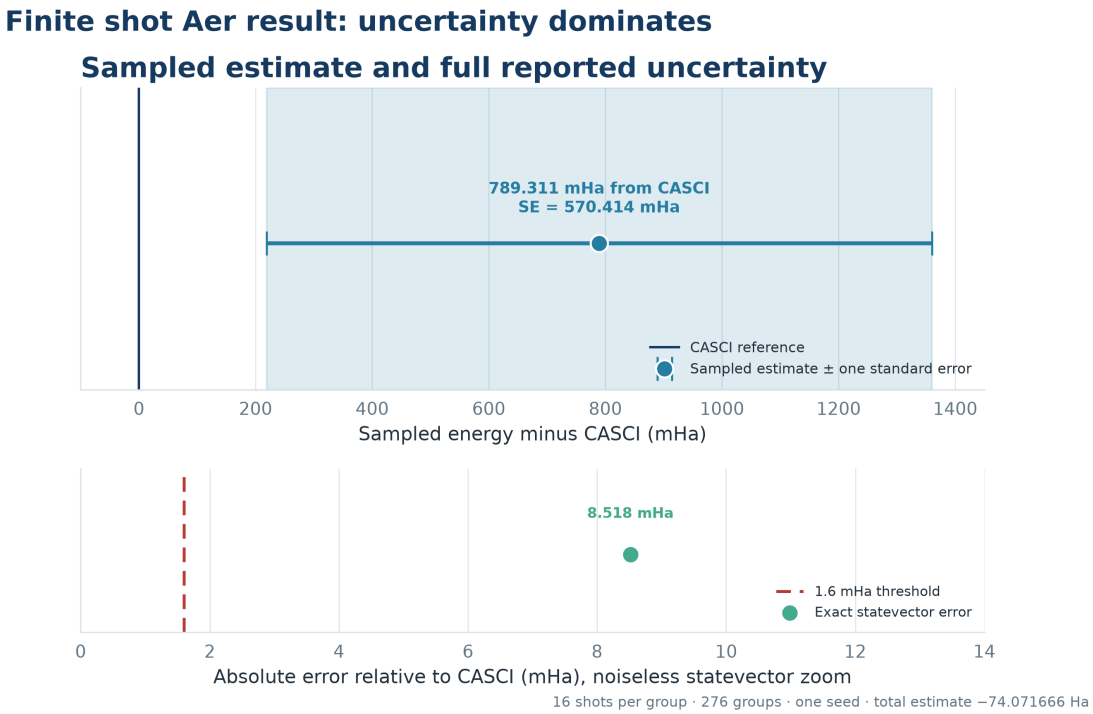


Figure 11: *Finite-shot Aer energy estimate and its sampling standard error.*

11 IBM hardware measurements

11.1 Hardware setup and measured state

The IBM jobs ran on `ibm_kingston`, a 156-qubit Heron r2 processor. The calibration timestamp recorded for the jobs was 2026 October 24 at 20:25 UTC. The software versions were Qiskit 1.4.3, qiskit-ibm-runtime 0.40.1, and qiskit-nature 0.7.2.

The molecular model was H_2O in the STO-3G basis at the main geometry. The active space was CAS(8e,6o) with the oxygen 1*s* core frozen, and the parity mapping with two-qubit reduction gave a ten-qubit register.

All four jobs measured the Hartree–Fock reference state. Its circuit uses four X gates and no entangling gates. We did not run the optimized PUCCSD state on the QPU. The three-repetition PUCCSD circuit has about 1,374 CX gates, and we report no hardware VQE energy for it.

The ideal energy of the prepared Hartree–Fock state is the RHF value, −74.623892 Ha. The CASCI reference is −74.860977 Ha, so the Hartree–Fock state sits about 237.085 mHa above CASCI by construction. Hardware measurements of this state should be judged against ideal Hartree–Fock. They are not optimized VQE energies.

11.2 Why the diagonal observable is exact for this reference

Under the mapping we use for state preparation, the Hartree–Fock state is a computational basis state. Any Pauli string containing an X or Y has zero expectation value in it. The ideal Hartree–Fock energy can therefore be computed from identity and Z-only terms:

$$\langle \Phi_{\mathrm{HF}} | \hat{H} | \Phi_{\mathrm{HF}} \rangle = E_{\mathrm{offset}} + \sum_{j \in \mathcal{Z}} c_j \langle \Phi_{\mathrm{HF}} | \hat{P}_j | \Phi_{\mathrm{HF}} \rangle, \tag{17}$$

where $\mathcal{Z}$ is the set of identity and Z-only terms. We checked in simulation that this restricted observable matches the full Hamiltonian expectation for the ideal Hartree–Fock state to within 10^{-9} Ha.

The shortcut belongs to this state. It is exact for the computational-basis Hartree–Fock state and cannot replace the full Hamiltonian when we estimate an entangled VQE state.

11.3 Job settings and results

Table 9: *IBM hardware measurements of the Hartree–Fock reference state.*

Job	Observable	Runtime configuration	Sampling record	Energy and reported standard error
A	Full Hamiltonian, 1,079 terms and 276 groups	Resilience level 1	Precision target 0.04 Ha, 104 seconds wall time	−73.967321 Ha; 346.52 mHa
B	79 identity and Z-only terms	Resilience level 1, TREX and twirling	40,000 shots, 32 randomizations, target 0.005 Ha not reached, about 21 seconds QPU time	−74.542299 Ha; 2,140.26 mHa
C	Same 79-term diagonal observable	Resilience level 0	100,000 shots requested	−74.327453 Ha; 3.45 mHa
D	Same diagonal observable on selected physical qubits	Resilience level 0	100,000 shots requested	−74.573420 Ha; 1.47 mHa

The offsets from ideal Hartree–Fock are:

Job B gave a point estimate closer to ideal Hartree–Fock than Job C, but its reported standard error was very large. Job C had a smaller standard error and a larger offset. The jobs used different configurations, so this comparison cannot isolate what error mitigation did.

Job D used the physical qubits 148, 42, 117, 149, 154, 44, 91, 134, 40, 89. Their reported mean readout error was 0.0031006, against a device mean of 0.0201588. In this one measurement, Job D landed closer to ideal Hartree–Fock than Job C did. It is a single job and is not evidence that this layout helps in general.

Table 10: *Measured energy offsets from the ideal Hartree–Fock energy.*

Job	Offset from ideal HF (mHa)	Interpretation
A	+656.571	Full Hamiltonian estimate with large statistical uncertainty
B	+81.593	Mitigated diagonal observable with very large reported uncertainty
C	+296.439	Unmitigated diagonal observable
D	+50.472	Unmitigated diagonal observable on selected qubits

We suspected readout error behind the large offset in Job C, but we did not verify that. Treat it as a possible source of systematic error and not as a confirmed diagnosis. The reported standard errors are statistical and leave out other systematic device effects.

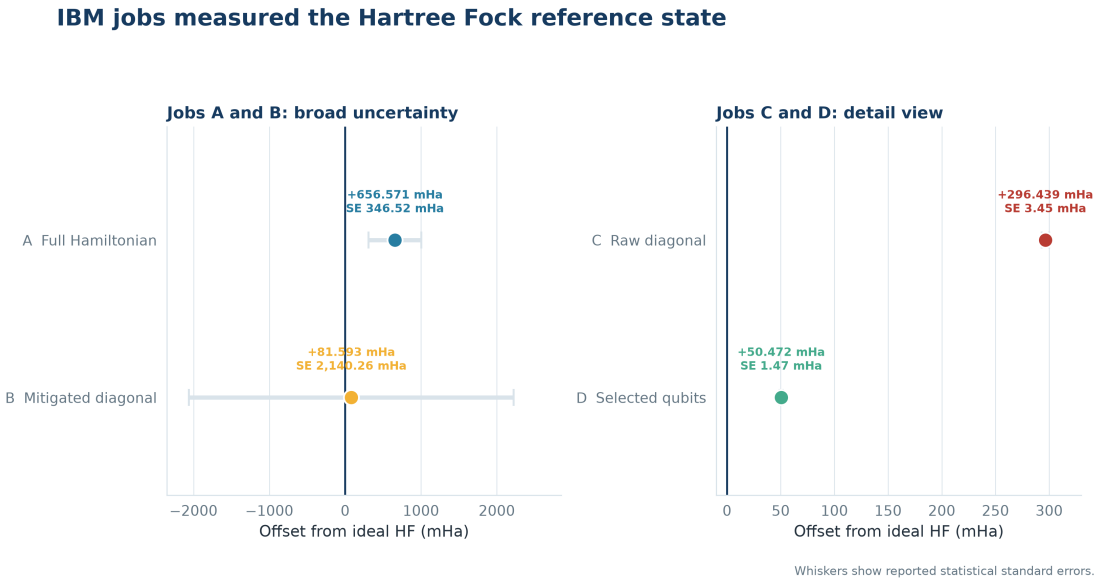


Figure 12: *IBM hardware measurements of the Hartree–Fock state, as offsets from ideal HF. These are not optimized VQE energies.*

11.4 Recorded IBM job identifiers

The recorded job identifiers are:

- Job A: `dbehvvfgdj9s73flrceg`

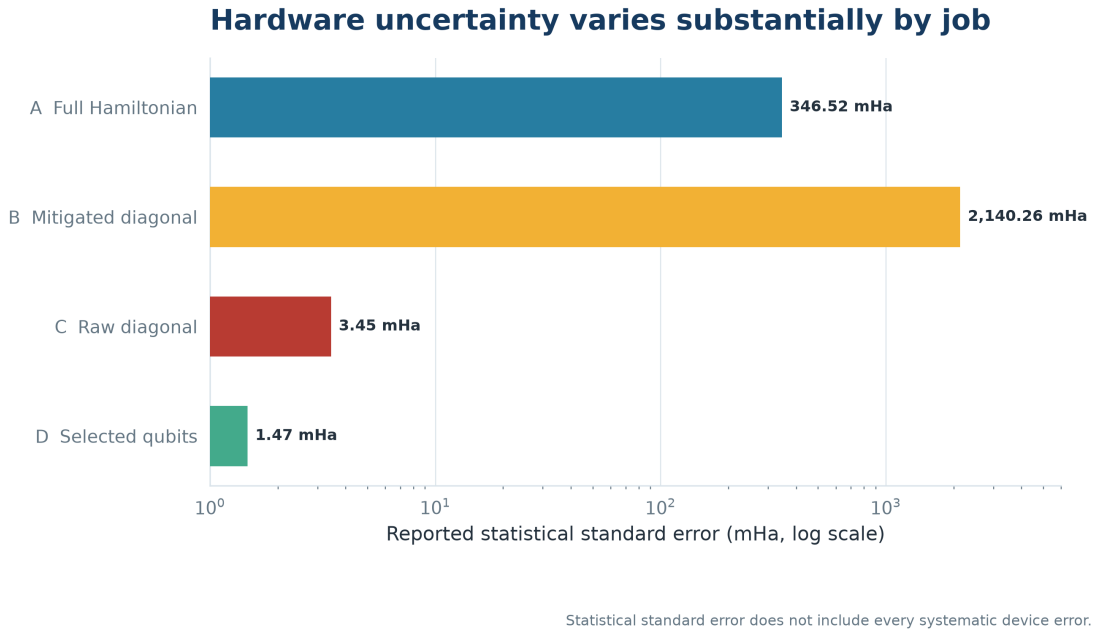


Figure 13: *Reported statistical standard errors for the four IBM jobs.*

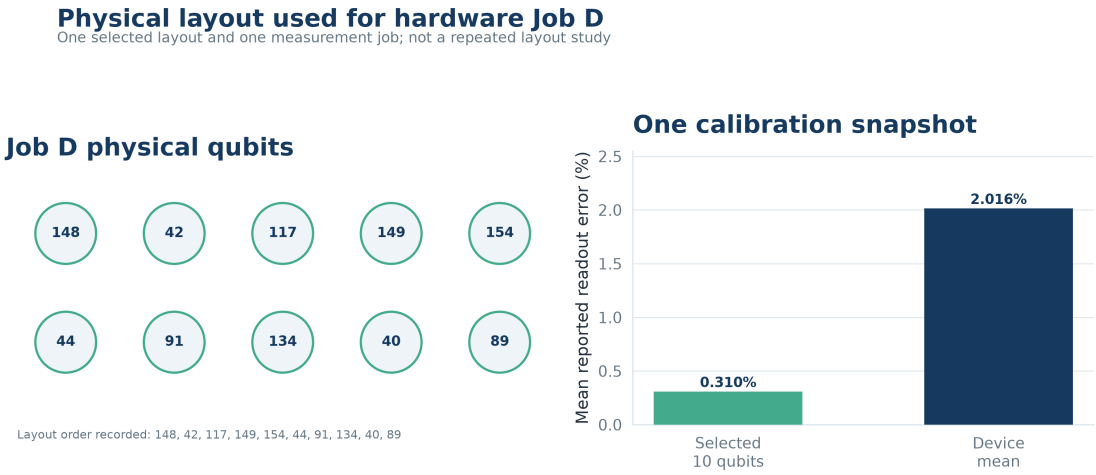


Figure 14: *Readout error summary for the selected Job D layout.*

- Job B: `dbei45vpc3tc739lomf0`
- Job C: `dbei5tvpc3tc739lop2g`
- Job D: `dbei8tom94ac738rsp90`

With these identifiers the jobs can be traced in the IBM Runtime interface. API keys, tokens, and other credentials must never go into the report.

11.5 What the hardware results show

The jobs show that we built and submitted a measurement pipeline for the Hartree–Fock reference state. They exercised Hamiltonian construction, mapping, transpilation, observable layout, Estimator V2 submission, and retrieval of results.

They do not give the energy of the optimized PUCCSD state, they do not show a hardware VQE optimization loop, and they do not show chemical accuracy. We report the hardware and noiseless results separately because they answer different questions.

12 Results viewed together

The project produced several kinds of evidence, and each should be read for the state and model that was actually measured.

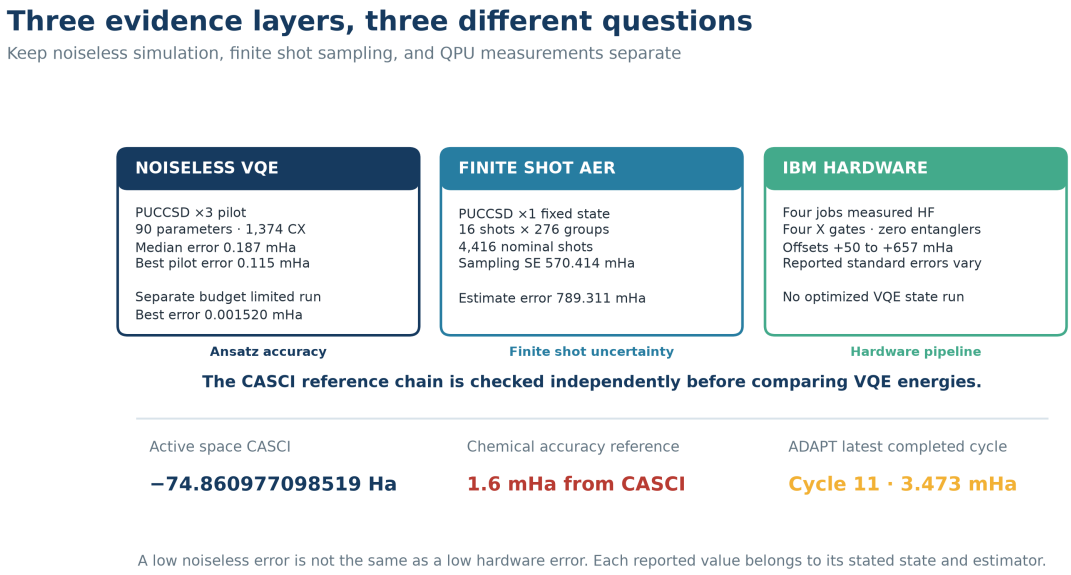


Figure 15: *Overview of project results, with each calculation type on its own scale.*

Table 11: *Summary of the main results and what each one supports.*

Result	Reported outcome	What it supports	What it does not support
Mapped exact check	Matches CASCI at reported precision	Validation of the active-space mapping chain	Accuracy beyond the chosen basis and active space
PUCCSD three-repetition resource pilot	Median error 0.187 mHa, best 0.115 mHa	High-accuracy noiseless simulation for this setup	Hardware accuracy or universal ansatz superiority
Three-trial PUCCSD L-BFGS-B run	Best error 0.001520 mHa; all trials reached budget	A high-accuracy observed parameter point	Optimizer convergence or global optimality
ADAPT run	Eleven completed cycles; final error 3.473 mHa	Adaptive circuit growth reduced the energy error	A completed sixteen-cycle result or chemical accuracy
Finite-shot Aer run	Error 789.311 mHa; standard error 570.414 mHa	A low-shot sampling demonstration	Useful chemical accuracy or a reliable energy estimate
IBM hardware jobs	Four measurements of the Hartree–Fock reference	A working hardware measurement pipeline	Hardware VQE or an optimized PUCCSD energy

13 What the team learned

Reference checks come first. CASCI and the exact mapped active-space calculation agree, which gives us confidence that the Hamiltonian and its qubit form describe the problem we meant to solve.

Ansatz size changes both accuracy and cost. In the resource pilot, more repetitions of the paired ansatz gave better noiseless energies, and they also gave more parameters and more CX gates. A circuit that does better in a noiseless simulation can be harder to use on a device, because two-qubit gates are where hardware errors pile up.

The optimization status has to be reported next to the energy. The best L-BFGS-B trial reached an excellent energy, but it used its whole evaluation budget. We are happy with the energy and we still label the run as budget-limited.

An adaptive algorithm can make steady progress and still miss the target. The ADAPT error dropped over eleven completed cycles and ended at 3.473 mHa, and the run did not finish the sixteen cycles we asked for.

Low-shot estimates can be dominated by statistical uncertainty. The finite-shot Aer result had an uncertainty hundreds of millihartree wide, and a point estimate quoted without it would look far more precise than it was.

Running something on hardware is not the same as running VQE on hardware. The IBM mea-

surements showed that a Hartree–Fock state and its observable can be executed through the service. We did not run the optimized PUCCSD circuit, so these jobs are not VQE ground-state results.

14 Limitations

1. **Minimal basis.** STO-3G is a minimal basis. The conclusions hold for this computational model and are not basis-set-converged predictions for water.

2. **Active space approximation.** The VQE calculations use CAS(8e,6o) with the oxygen 1*s* core frozen. VQE errors are measured against this active-space CASCI energy.

3. **Small ansatz pilot.** The resource pilot used two runs per configuration. Its medians and best values describe those runs only.

4. **Budget-limited optimization.** The three-trial PUCCSD L-BFGS-B run reached 1,500 objective evaluations in every trial. The output does not show convergence.

5. **Sparse geometry scan.** The warm-start scan has three geometries. It shows the procedure but does not define a continuous potential energy curve.

6. **Incomplete ADAPT run.** The later ADAPT calculation asked for sixteen macrocycles, completed eleven, and stopped as cycle twelve began. The pasted log has no completed result for cycles twelve through sixteen.

7. **Low-shot result.** The Aer calculation used sixteen shots per group and one seed. Its standard error is far above the chemical-accuracy threshold.

8. **Noise configuration reporting.** The pasted Aer output gives the completed estimate and the shot budget, but not every simulator noise-channel detail. We do not assign a device-calibrated noise model without checking the saved run configuration.

9. **Hardware state.** All four IBM jobs prepared the Hartree–Fock reference state. The optimized PUCCSD circuit was not measured on the QPU.

10. **Hardware uncertainty.** The reported standard errors are statistical. They leave out systematic sources such as readout bias, drift, leakage, and crosstalk.

11. **Hardware comparison.** The four jobs used different settings and are individual observations. They are not a controlled, repeated benchmark of mitigation methods.

15 Conclusion

We built and checked a VQE workflow for asymmetric stretching of water in a defined STO-3G active space. The exact solve of the mapped active-space problem agrees with CASCI. In noiseless simulation, the three-repetition paired ansatz reached sub-millihartree errors in the reported pilot. One separate three-trial L-BFGS-B run gave a best difference of 0.001520 mHa from CASCI, although every trial stopped at its evaluation limit.

The adaptive run brought the error down to 3.473 mHa after eleven completed cycles and was interrupted before the sixteen requested cycles finished. The low-shot Aer calculation completed, but its 570.414 mHa standard error rules it out as an accurate energy estimate. The IBM measurements ran the pipeline for the Hartree–Fock reference state and did not run the optimized ansatz.

The strongest conclusion we can support is a specific one. For this active-space water problem, the tested paired ansatz represents the ground state accurately in noiseless simulation. The finite-shot and IBM experiments show separate measurement challenges, and they do not establish chemical accuracy for VQE on hardware.

Acknowledgements

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